

Home Assignment

Department: Computer Science Engineering: AI & ML

Subject: Discrete Mathematic

Date: 06-11-2025

Year: S Y

Sem: I

A Y: 2025-26

Max Marks: 30

Last Date of Submission: 14-11-2025

- Q.1) Express the statements “Some student in this class has visited Mexico” and “Every student in this class has visited either Canada or Mexico” using predicates and quantifiers.
- Q.2) The logician Raymond Smullyan describes an island containing two types of people: knights who always tell the truth and knaves who always lie. You visit the island and are approached by two natives who speak to you as follows:

A says: *B* is a knight.

B says: *A* and I are of opposite type.

What are *A* and *B*?

- Q.3) Write a formal and an informal contrapositive, converse, and inverse for the following statement:

If a real number is greater than 2, then its square is greater than 4.

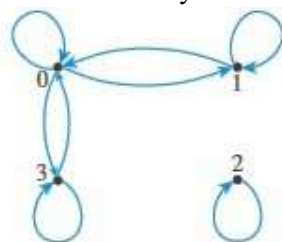
- Q.4) Among the integers 1 to 1000
- a) How many of them are not divisible by 3, nor by 5, nor by 7?
 - b) How many are not divisible by 5 and 7 but not by 3
 - c) How many are divisible by 2,3,5 or 7
- Q.5) Formulate and prove by induction, a general formula stemming from the observation
- a) $1^3 = 1$
 - b) $2^3 = 3 + 5$
 - c) $3^3 = 7 + 9 + 11$
 - d) $4^3 = 13 + 15 + 17 + 19$
- Q.6) Let $A = \{0, 1, 2, 3\}$ and define relations R , S , and T on A as follows:

$R = \{(0, 0), (0, 1), (0, 3), (1, 0), (1, 1), (2, 2), (3, 0), (3, 3)\}$,

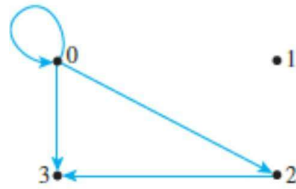
$S = \{(0, 0), (0, 2), (0, 3), (2, 3)\}$,

$T = \{(0, 1), (2, 3)\}$.

- a. Is R reflexive? symmetric? transitive?



b. Is S reflexive? symmetric? transitive?



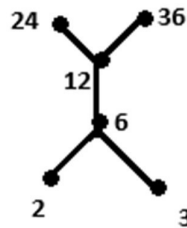
c. Is T reflexive? symmetric? transitive?



Q.7) Find the transitive closures of R by Warshall's Algorithm, where $A = \{1, 2, 3, 4, 5, 6\}$ and $R = \{(x, y) \mid |x - y| = 2\}$.

Q.8) Let $x = \{2, 3, 6, 12, 24, 36\}$ such that $x < y$ if x divides y then Find the following

- a) Maximal Element
- b) Minimal Element
- c) Chain
- d) antichain
- e) The given POSET in figure is a Lattice or Not.



Q.9) Let Set $A = B$ be the set of Real Numbers

$F: A \rightarrow B$ given by $f(x) = 2x^3 - 1$

$G: B \rightarrow A$ given by $g(y) = \sqrt[3]{\frac{1}{2}y + \frac{1}{2}}$

Show that f is a bijective function and g is also bijective function

Q.10) Let $f(x) = x + 2$, $g(x) = x - 2$, $h(x) = 3x$ for x belongs to R , where R is a set of real numbers, then find the following

gof , fog , fof , gog , foh , hog , hpf , and fohog

Q.11) A certain computer access password consists of 3 through 5 uppercase letters chosen from the 26 letters in the Roman alphabet, with repetitions allowed.

- a. How many different passwords are possible?
- b. How many different passwords have no repeated letter?
- c. How many different passwords have at least one repeated letter?
- d. If all passwords are equally likely, what is the probability that a randomly chosen password has at least one repeated letter?

Q.12) Suppose the group of twelve consists of five men and seven women.

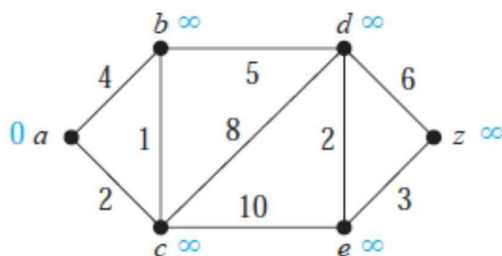
- a. How many five-person teams can be chosen that consist of three men and two women?
- b. How many five-person teams contain at least one man? Solve by addition rule and difference rule
- c. How many five-person teams contain at most one man?

Q.13) Prove the Binomial Theorem (algebraic version) through mathematical Induction

Q.14) Solve the congruence equation $1092x \equiv 213 \pmod{2295}$

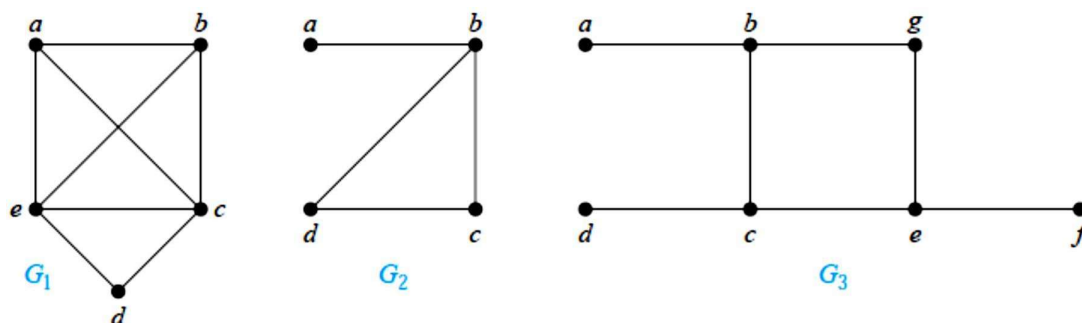
- Q.15) Solve the following by using Chinese Remainder Theorem with method 1 and method 2,
- $x \equiv 2 \pmod{3}$ and
 - $x \equiv 4 \pmod{5}$
 - $x \equiv 14 \pmod{15}$ and
 - $x \equiv 6 \pmod{7}$

Q.16



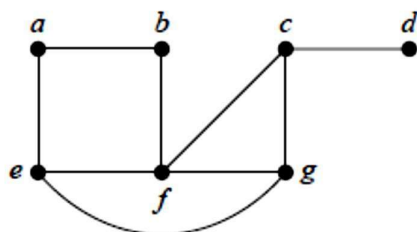
For the graph given find out the shortest path using step by step using Dijkstra's Algorithm

Q.17) Which of the simple graphs in Figure have a Hamilton circuit or, if not, a Hamilton path?



Q.18) Suppose that someone starts a chain letter. Each person who receives the letter is asked to send it on to four other people. Some people do this, but others do not send any letters. How many people have seen the letter, including the first person, if no one receives more than one letter and if the chain letter ends after there have been 100 people who read it but did not send it out? How many people sent out the letter? Solve by using Graph Tree

Q.19) Find a spanning tree of the simple graph G shown in Figure



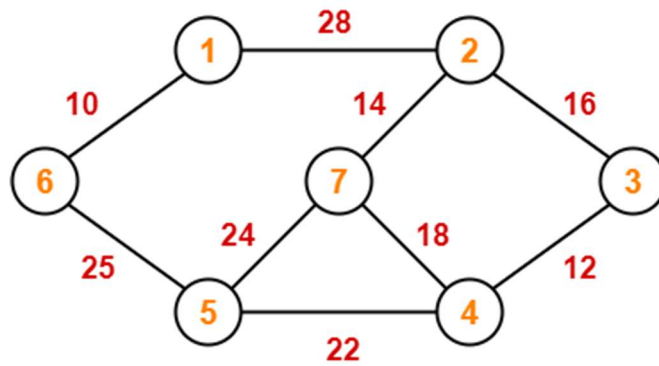
Q.20) Go through the steps to build a binary search tree for the keys 15, 10, 19, 25, 12, 4, and insert the keys in the order in which they are listed. For simplicity, use the same names for the vertices and their associated keys. Show the sequence of steps in tree diagrams.

Q.21) For the above binary search tree, delete the nodes consisting of the keys 12 and 19. Also draw the resultant tree. Write all the algorithms of node deletion with 3 cases.

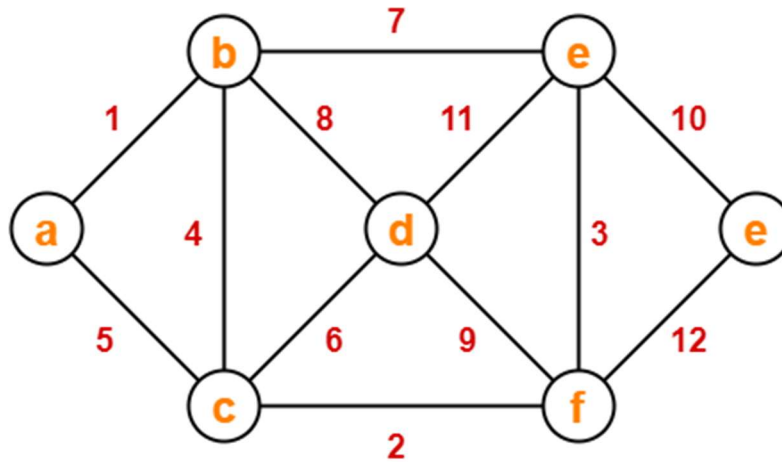
Q.22) Create Huffman tree with prefix codes for the following letters given with frequencies.

C	D	E	K	L	M	U	Z
35	43	123	9	43	24	39	5

Q.23 Use Prim's Algorithm and find MST. Solve it step by step.



Q.24 Use Kruskal's Algorithm and find MST. Solve it step by step.



Q.25. In how many ways a committee consisting of 5 men and 3 women, can be chosen from 9 men and 12 women?

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